Data Sources and Provenance

Female Unemployment Across Los Angeles County Census Tracts
American Community Survey 2020–2024 Five-Year Estimates

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August 26, 2026

1 Overview

All data in this project come from two public U.S. Census Bureau sources: the **American Community Survey (ACS) 2020–2024 five-year estimates**, distributed as table-based summary files, and the **TIGER/Line 2024** census-tract boundary shapefiles. No third-party or proprietary data are used, and no API key is required for any download.

The 2020–2024 five-year release (published December 2025) is the most recent ACS release carrying census-tract-level detail. Estimates are five-year averages over the 2020–2024 collection period, not a 2024 snapshot.

2 Why the summary files rather than the Census API

The conventional route to ACS data, the `api.census.gov` data endpoints, now requires a (free) API key. This pipeline instead reads the public *table-based summary files* — static pipe-delimited downloads hosted on `www2.census.gov` that need no key or registration:

`https://www2.census.gov/programs-surveys/acs/summary_file/2024/table-based-SF/data/5YRData/`

Each table is a single national file (roughly 80–305 MB, one row per geography at every published summary level). The build script (`R/01_build_data.R`) downloads each national file to a temporary location, keeps only the rows whose `GEO_ID` begins with `1400000US06037` — summary level 140 (census tract), state 06 (California), county 037 (Los Angeles) — writes that slice to `data/raw/`, and deletes the national file. The SAS reproduction (`sas/la_female_unemployment.sas`) follows the same procedure and includes a keyed API alternative in a trailing comment.

3 ACS tables used

Five detailed tables were retrieved, each from a URL of the form `acsdt5y2024-`*`table`*`.dat` under the directory above.

A note on B23001 versus the obvious shortcut: subject table S2301 publishes a ready-made female unemployment rate, but only for ages 20–64. Building the measure from B23001 covers

Table	Subject	Used for
B23001	Sex by age by employment status	Female and male unemployment rates (the outcome measure) and their margins of error
B19013	Median household income	Income correlate; income-quintile gradient
B17001	Poverty status by sex by age	Tract poverty share
B15003	Educational attainment (25+)	Share with bachelor's degree or higher
B03002	Hispanic or Latino origin by race	Total population; Hispanic, non-Hispanic White, Black, and Asian shares

Table 1: ACS 2020–2024 five-year detailed tables. The Los Angeles County slices are retained in `data/raw/` (2,498 tract rows each).

the full 16-and-over civilian labor force and allows margins of error to be aggregated properly from the component cells.

4 Tract geometry

Census-tract boundaries come from the TIGER/Line 2024 shapefile for California:

`https://www2.census.gov/geo/tiger/TIGER2024/TRACT/tl_2024_06_tract.zip`

The build script keeps the Los Angeles County tracts (`COUNTYFP == 037`), retains the `GEOID`, `NAMELSAD`, `ALAND`, and `AWATER` fields, transforms to NAD83 (EPSG:4269), and simplifies geometry with a tolerance of 2×10^{-4} degrees (topology preserved). The result is saved as `data/la_tracts_geom.rds`; a further-simplified version is exported to `docs/data/tracts.json` for the static site.

5 Processing applied to the source data

- **Sentinel values.** The summary files publish -666666666 and related sentinels for "not available"; these are recoded to missing.
- **Outcome measure.** Female unemployment rate = women 16+ unemployed ÷ women 16+ in the *civilian* labor force, summed over every age band of B23001. For ages 16–64 the civilian labor force is read from the `Civilian:` lines (the labor force there splits into Armed Forces and Civilian); for ages 65+ there is no Armed Forces line, so `In labor force:` is itself the civilian total.
- **Margins of error.** Component MOEs are aggregated by the rules of the ACS General Handbook, chapter 8: root sum of squares for sums, counting only the single largest zero-estimate cell; the proportion formula for the rate, falling back to the ratio formula where the radicand is negative. Published MOEs are at the 90% confidence level, so standard errors are MOE/1.645.
- **Reliability.** A coefficient of variation is carried for every tract. Tracts whose estimated

rate is exactly 0% (70 tracts) have an undefined CV and are left missing; tracts with no women in the civilian labor force (25 tracts) are suppressed. Of the analyzable tracts, 84% have CV > 40% — "unreliable" by Census Bureau guidance — so individual tract values should be read with their confidence intervals, not as precise numbers.

6 Source references

- ACS 5-Year Data (2009–2024), Census developer documentation: `https://www.census.gov/data/developers/data-sets/acs-5year.html`
- ACS table-based summary files, 2024 release: `https://www2.census.gov/programs-surveys/acs/summary_file/2024/table-based-SF/`
- TIGER/Line 2024 census tract shapefiles: `https://www2.census.gov/geo/tiger/TIGER2024/TRACT/`
- ACS General Handbook, ch. 8 (calculating measures of error for derived estimates): `https://www.census.gov/content/dam/Census/library/publications/2018/acs/acs_general_handbook_2018_ch08.pdf`
- ACS General Handbook, ch. 7 (understanding error and determining statistical significance): `https://www.census.gov/content/dam/Census/library/publications/2018/acs/acs_general_handbook_2018_ch07.pdf`
- Census Bureau guidance for labor force statistics data users: `https://www.census.gov/topics/employment/labor-force/guidance.html`

Repository: `https://github.com/joelawalsh01/la-female-unemployment` · Live site: `https://joelawalsh01.github.io/la-female-unemployment/`
Data retrieved July 29, 2026 by `R/01_build_data.R`.